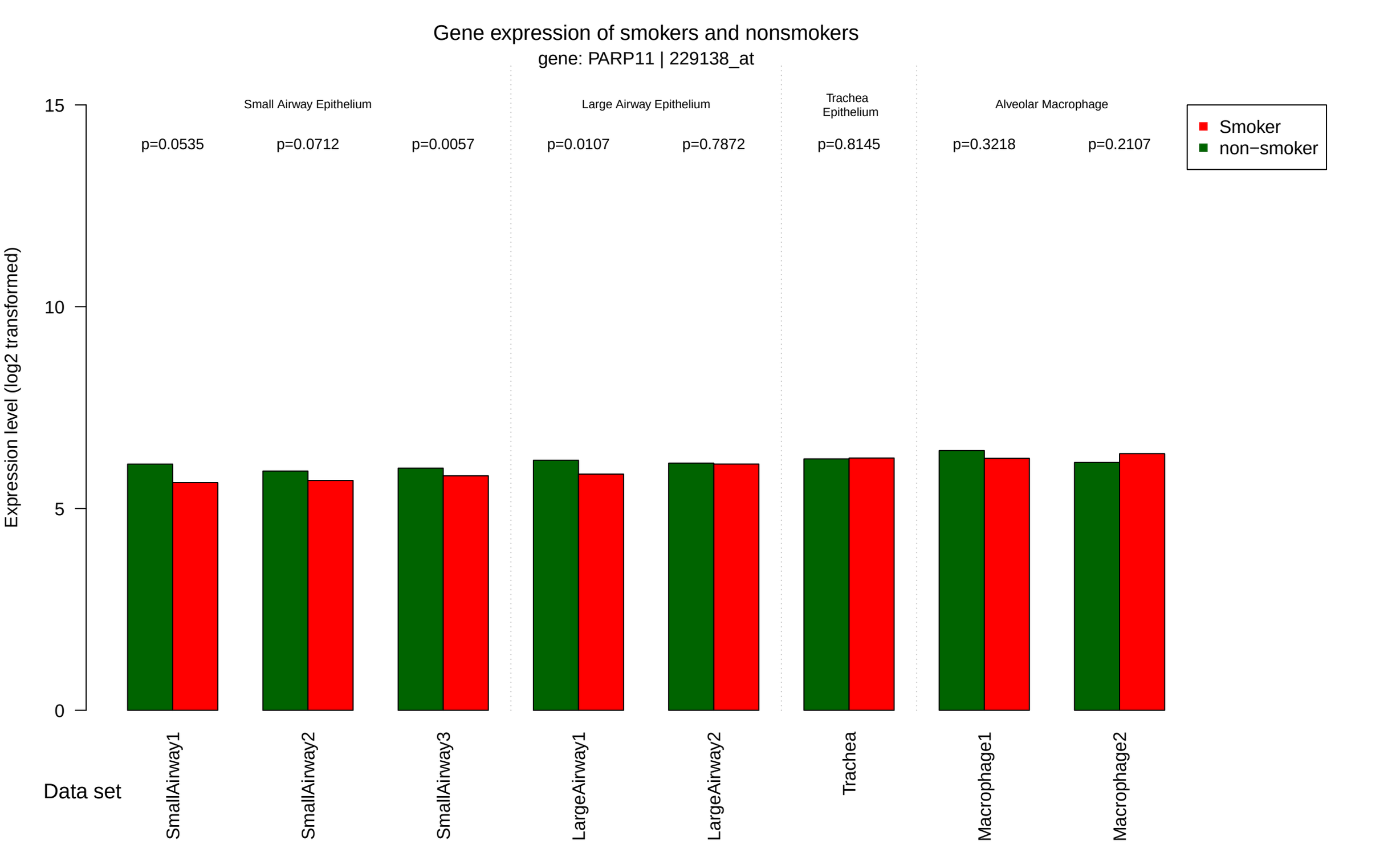
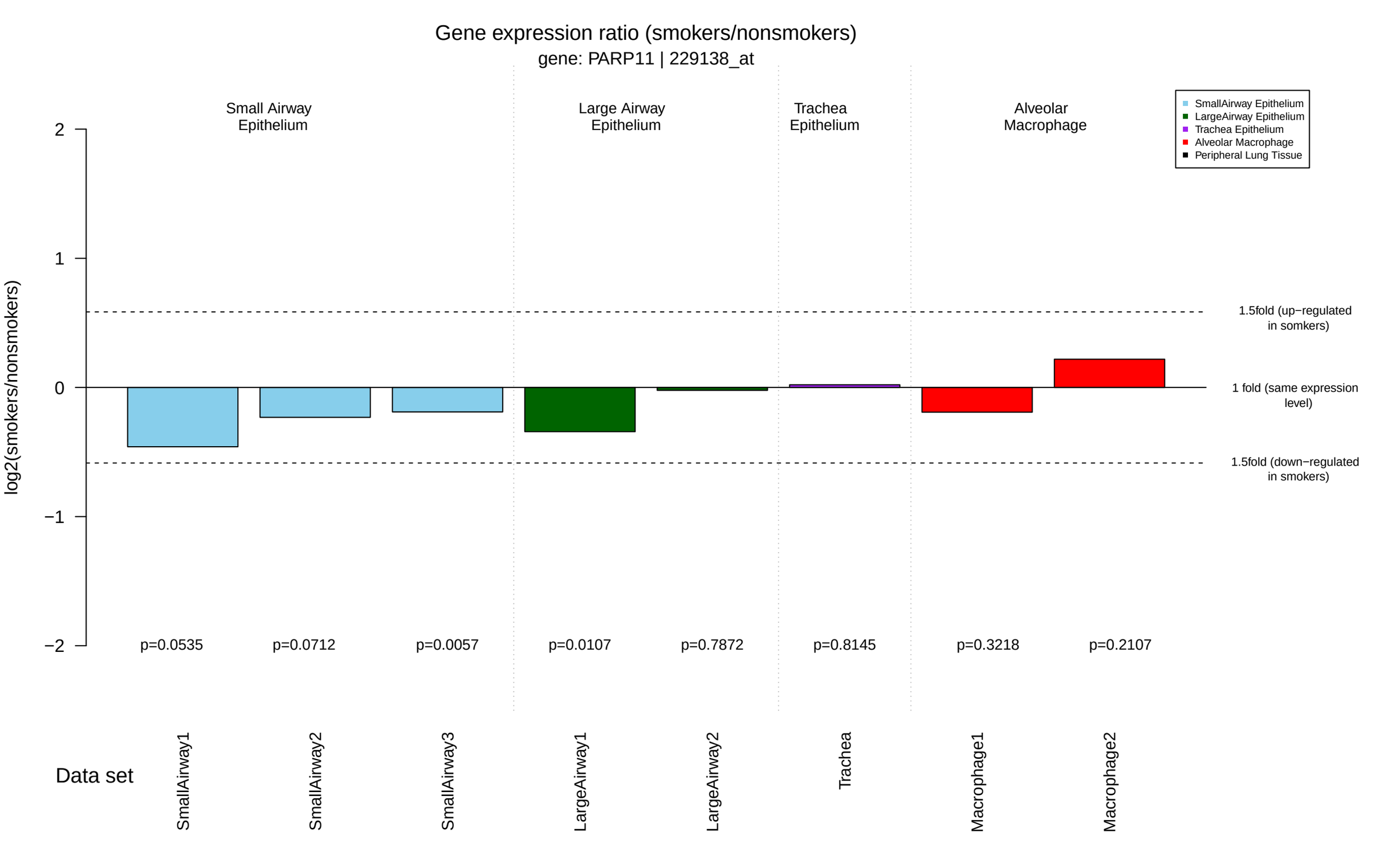
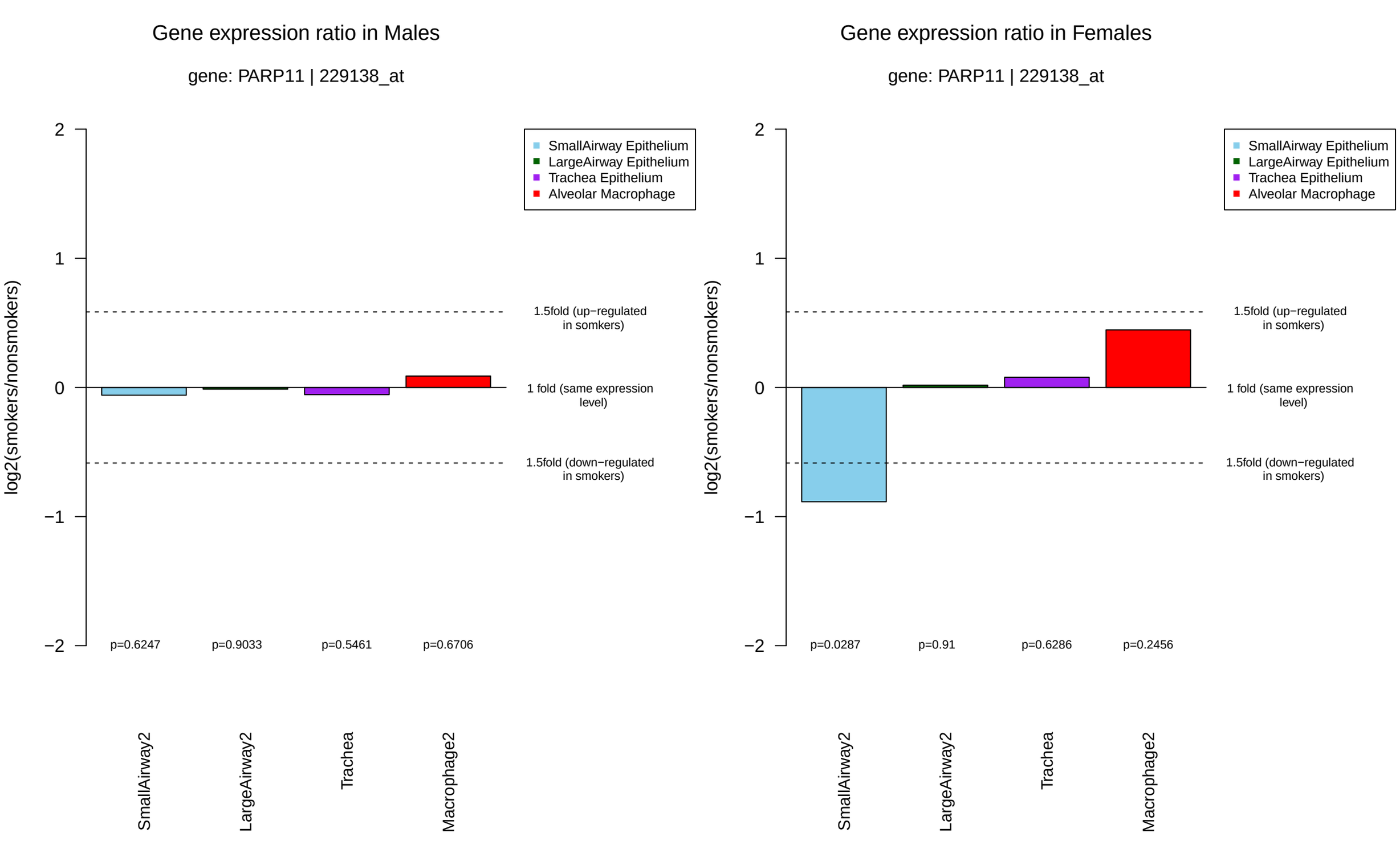
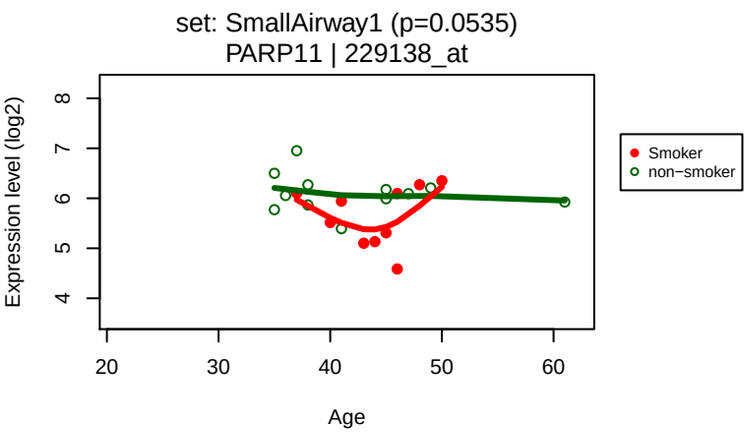
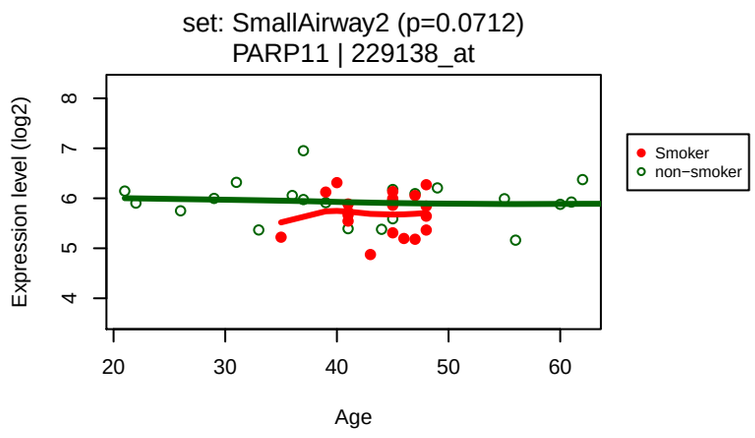
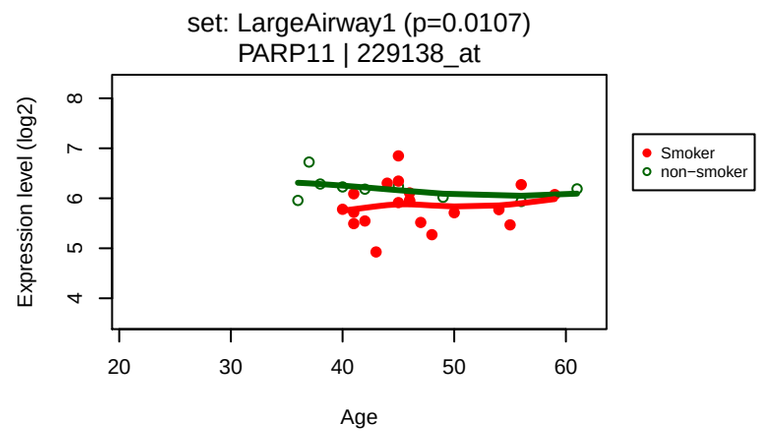

Gene expression of smokers and nonsmokers
gene: PARP11 | 229138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0535
p=0.0712
p=0.0057
p=0.0107
p=0.7872
p=0.8145
p=0.3218
p=0.2107
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

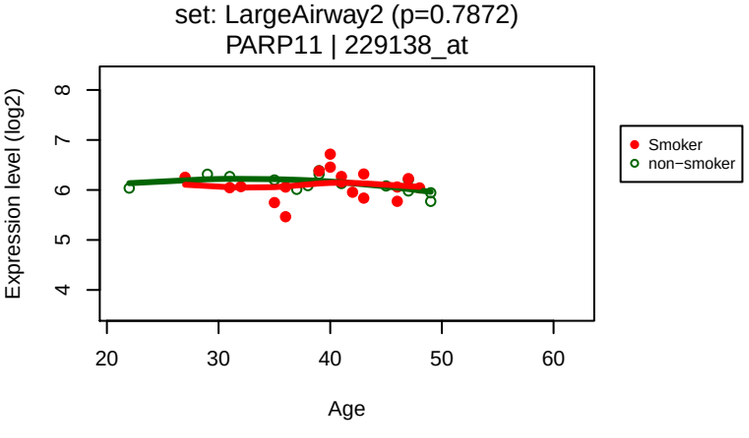

Gene expression ratio (smokers/nonsmokers)
gene: PARP11 | 229138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0535
p=0.0712
p=0.0057
p=0.0107
p=0.7872
p=0.8145
p=0.3218
p=0.2107
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

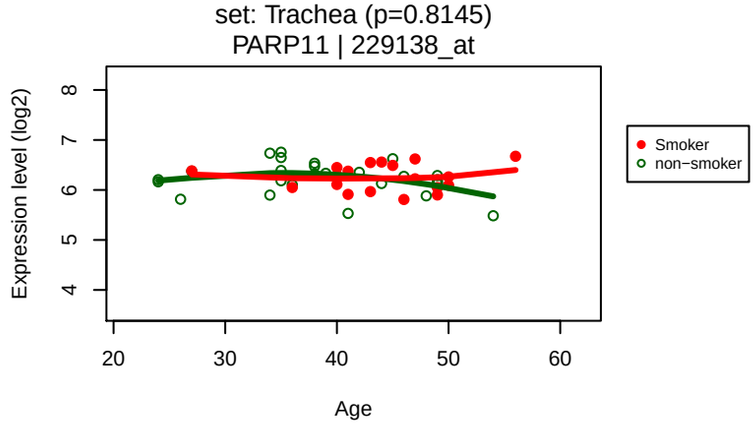

Gene expression ratio in Males
gene: PARP11 | 229138_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6247
p=0.9033
p=0.5461
p=0.6706
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PARP11 | 229138_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0287
p=0.91
p=0.6286
p=0.2456
SmallAirway2
LargeAirway2
Trachea
Macrophage2

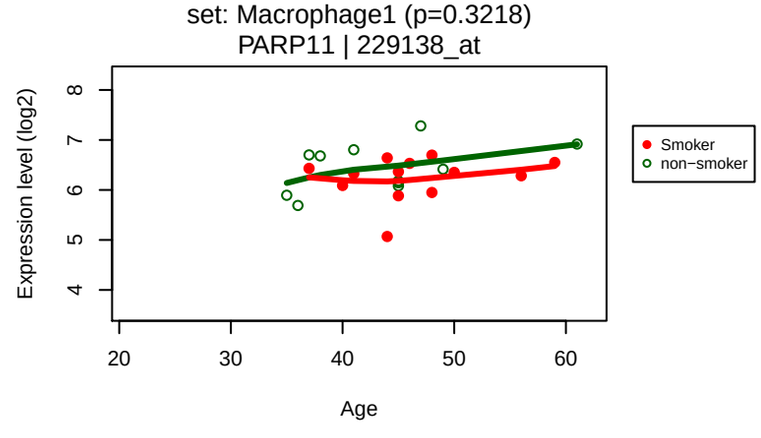

set: SmallAirway1 (p=0.0535)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker

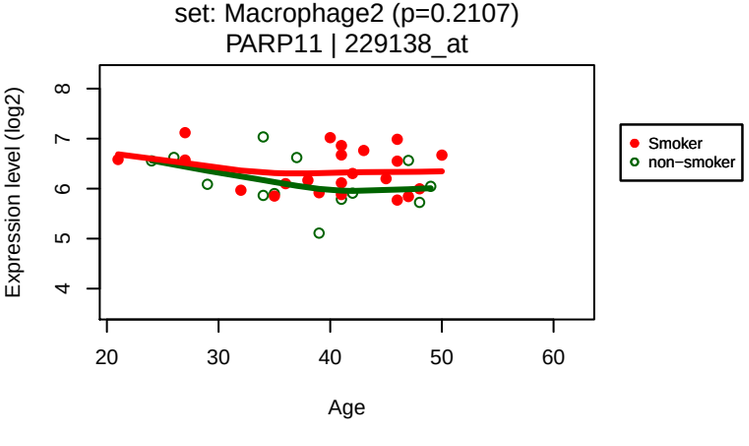

set: SmallAirway2 (p=0.0712)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker

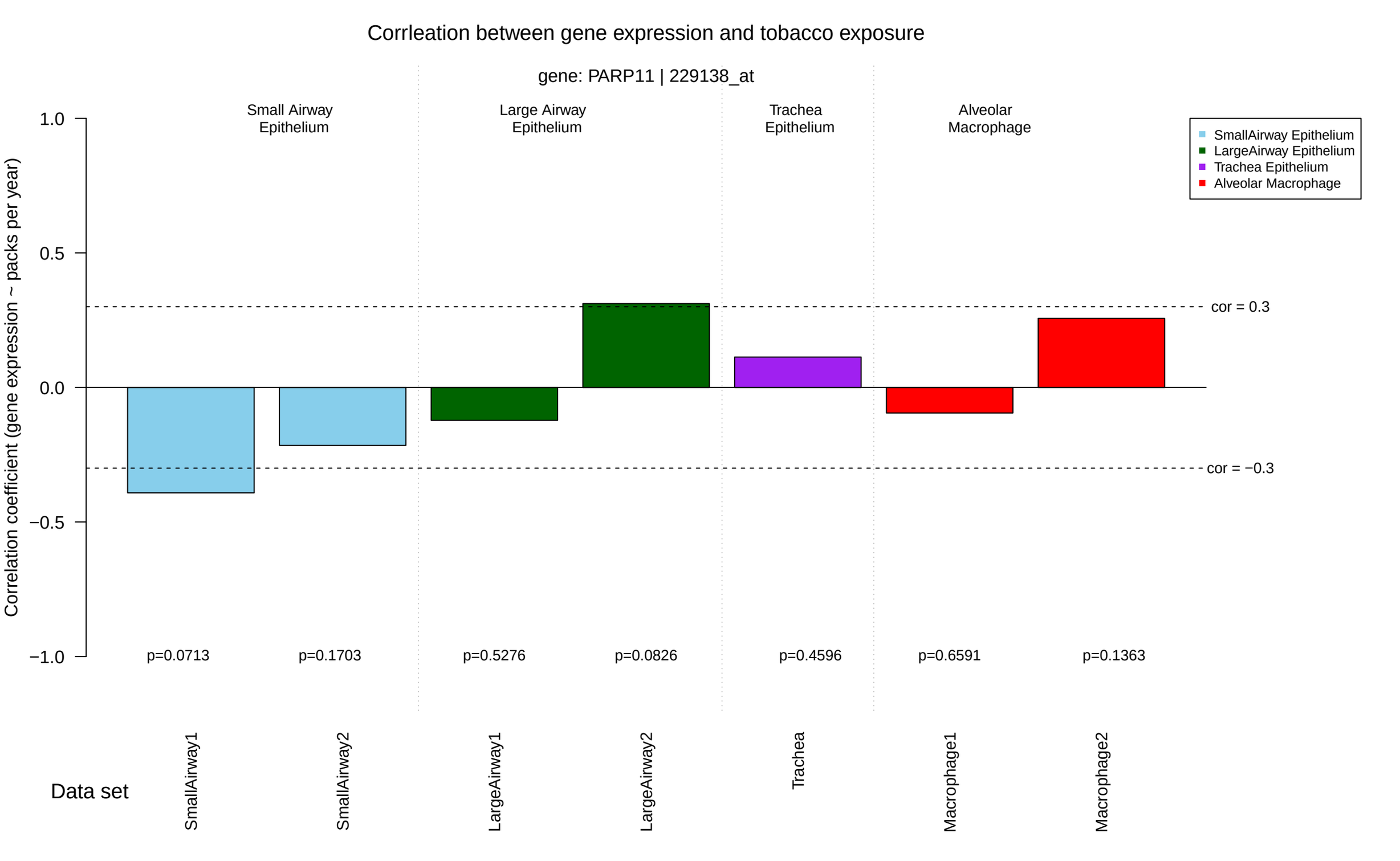

set: LargeAirway1 (p=0.0107)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7872)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8145)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3218)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2107)
PARP11 | 229138_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PARP11 | 229138_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0713
p=0.1703
p=0.5276
p=0.0826
p=0.4596
p=0.6591
p=0.1363
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2